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Frontier Sovereign Bond Returns in Dollar Terms: Carry, Currency, and Default Risk

Kilambi Research

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Abstract. We rebuild frontier-market sovereign bond returns in US dollar terms, with explicit default adjustments, and test whether ex-ante variables predict subsequent returns. The sample covers 18 frontier and emerging-market economies, 2010 to 2024, using 10-year local-currency sovereign yields. Three results emerge. First, the carry that dominated local-currency rankings does not survive the currency leg: Ghana, the 267 percent local-currency poster child of KR-WP-2026-03 (measured over 2016 to 2026), earned minus 78 percent in dollar terms over 2010 to 2024 once cedi depreciation and the 2023 domestic debt exchange are accounted for (Modelled, B). Second, the starting yield still predicts forward dollar returns, and the predictive power strengthens with horizon: one percentage point of additional starting yield is associated with about 2.4 percentage points of additional 12-month dollar return and about 7.8 percentage points of additional 60-month dollar return (with-controls specification at 60 months: b=0.078, t=2.02, R2=0.36, n=101, country-clustered standard errors). Third, the mechanism is the currency wedge: higher starting yields predict subsequent currency depreciation (correlation +0.245), and local-currency carry is strong before the FX leg eats most of it (t=2.89 local vs t=1.46 in dollars at 36 months). Rising debt-to-GDP ratios predict lower dollar returns, while the IMF program coefficient is negative but statistically insignificant at all horizons. Three falsifiers qualify the findings: the carry premium is a pre-2016 phenomenon (2016 to 2021 formations: b=0.006, R2=0.01), it is sensitive to the inclusion of Moldova, and Ghana drags the full-sample slope downward.

Keywords: frontier sovereign bonds, carry, currency risk, sovereign default, return predictability

JEL Classification: G12, G15, F34

Production note. This paper was produced by Kilambi Research in October 2026. Computations were performed in Python on the archived series inventoried in Appendix A; figures were rendered with Kilambi Research's institutional chart style. Data were retrieved in October 2026 and are subject to revision by the underlying sources. All analysis was performed by Kilambi Research's quantitative research pipeline with AI assistance and verified against source data.

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1 Introduction

A dollar investor who buys a frontier-market government bond in local currency earns the bond's yield, minus the currency's depreciation, minus whatever a restructuring takes. The accounting identity is simple; the economics is not. Frontier sovereigns offer the highest nominal yields in the fixed-income universe. Whether those yields compensate for the depreciation and default risk that accompany them is the question this paper asks.

The motivation is a quarantined result. KR-WP-2026-03 reported that frontier-market carry topped its 2016 to 2026 modeled 10-year buy-and-hold rankings in local-currency terms: Ghana at plus 267 percent, Kyrgyzstan at plus 192 percent. Those figures were flagged as indicative, not conclusions: they had no FX leg, unaligned windows, no archived build script, and Ghana's number ignored the cedi collapse and the 2023 domestic debt restructuring. A local-currency ranking that ignores the currency is a ranking of the wrong object for any investor who spends dollars. This paper rebuilds the exercise in dollar terms with explicit default adjustments, then tests whether ex-ante variables predict subsequent returns. It is a general study of frontier sovereign return predictability, not a Ghana case study.

We assemble a monthly panel for 18 frontier and emerging-market economies, 2010 to 2024, from 10-year local-currency sovereign yields and spot exchange rates, with haircuts applied only where local-currency debt was actually restructured. From it we compute dollar total returns via a duration approximation (Modelled, B), form December portfolios, and run predictive regressions of forward dollar returns on the starting yield and controls at 12, 36 and 60 month horizons, with country-clustered standard errors.

Four results emerge. First, carry does not survive the currency leg as a buy-and-hold proposition. Ghana, the plus 267 percent local-currency poster child, earned minus 78 percent in dollar terms over 2010 to 2024 (Modelled, B); Kyrgyzstan, the other poster child, earned minus 13 percent. The two economies that looked best in local currency sit at the bottom in dollars. Second, the starting yield still predicts forward dollar returns, and the predictive power strengthens with horizon: the with-controls slope rises from 0.024 at 12 months (t=1.33, R2=0.13, n=153) to 0.059 at 36 months (t=1.92, R2=0.30, n=129) to 0.078 at 60 months (t=2.02, R2=0.36, n=101). Tercile sorts tell the same story: the high-minus-low yield tercile earned plus 10.9 percentage points at 12 months, plus 28.9 at 36 months, and plus 35.7 at 60 months. Third, the mechanism is the currency wedge. Higher starting yields predict subsequent currency depreciation (correlation +0.245; slope 0.011, t=1.99), and carry is strong in local currency (t=2.89 at 36 months) but mostly offset by the FX leg in dollars (t=1.46). Fourth, the "ruin" leg is real in the multivariate results, but it runs through debt accumulation rather than IMF programs: a rising debt-to-GDP ratio predicts lower 60-month dollar returns (b=-0.018, t=-2.57), while an active IMF program, the distress signal, carries a negative but statistically insignificant coefficient at every horizon (60 months: b=-0.30, t=-1.22, p=0.22).

We report three falsifiers prominently because they bound what the results can claim. The carry premium is a pre-2016 phenomenon: formations from 2016 to 2021 show a slope of 0.006 with R2=0.01 at 36 months, against 0.057 with R2=0.23 for 2010 to 2015. The premium is sensitive to Moldova: dropping it halves the 36-month slope to 0.016 (though the t-statistic rises to 2.75). And Ghana drags the full-sample slope downward: dropping Ghana raises the 36-month slope to 0.068. The default adjustments, by contrast, do not drive the premium: removing the haircuts leaves the 36-month slope essentially unchanged (0.0366 vs 0.0363).

The rest of the paper proceeds as follows. Section 2 describes how frontier sovereign bond markets

price carry, currency and default risk. Section 3 reviews the related literature. Section 4 documents the data. Section 5 sets out the methodology. Section 6 presents the results. Section 7 discusses falsifiers, limitations and falsifiability. Section 8 concludes. Appendix A inventories the data.

2 Institutional background

A dollar-based investor who buys a frontier-market sovereign bond in local currency takes three exposures at once. The first is the bond itself: a 10-year promise by the sovereign to pay coupons and principal in its own currency. The second is the currency: every coupon and the principal must be converted back to dollars at future spot rates. The third is restructuring risk: the sovereign may change the terms.

The dollar total return decomposes accordingly. To first order,

$$R_{USD} \approx R_{local} + R_{FX}, \tag{1}$$

where R_{local} is the bond's total return in local currency and R_{FX} is the currency's return against the dollar (negative when the local currency depreciates). Expected dollar return is therefore approximately the starting yield, minus expected depreciation, minus expected default loss, plus expected price appreciation from yield changes. This paper's core question is whether the starting yield survives those subtractions as a predictor of realized returns.

The carry trade in currencies is the direct analogue: borrow in low-yield currencies, lend in high-yield ones, and pocket the rate differential. Uncovered interest parity says the differential should be offset by depreciation; in practice it often is not, at least over some horizons (Lustig et al., 2011; Menkhoff et al., 2012). Frontier sovereign bonds are the fixed-income version of the same bet, with duration added. The appeal is the starting yield: frontier 10-year yields in our sample average about 8 percent against roughly 2 to 4 percent for core markets over the same period. The risk is that the yield is the market's forecast of exactly the depreciation and restructuring that will erase it.

Hedging the currency leg is usually not a practical alternative for these markets. Forward markets in frontier currencies are thin or absent, and where forwards exist, covered interest parity deviations make hedging expensive (Du et al., 2018). The unhedged dollar return is therefore the economically relevant object, and it is what we measure.

Restructurings need a precise treatment because external and domestic debt are different contracts. When a sovereign defaults on external, foreign-law bonds, a holder of local-currency, local-law bonds is not directly haircut; the damage arrives through the currency collapse and the yield spike that accompany the crisis. When the sovereign restructures domestic, local-law debt, as Ghana did in its February 2023 Domestic Debt Exchange Programme (DDEP) and Sri Lanka did in its September 2023 Domestic Debt Optimisation, the local bondholder takes a direct net-present-value loss. Our panel applies haircuts only in the second case: a 14 percent NPV reduction for Ghana in February 2023 (the IMF's central figure in Country Report 23/168; (Modelled, C), given the wide analyst range) and a 10 percent NPV judgment for Sri Lanka in September 2023 (Modelled, C). External-only defaults enter through observed FX and yield moves, which is the correct economics for a local-currency bondholder. Cruces and Trebesch (2013) document the wide dispersion of haircuts across restructuring episodes, which is why we grade these adjustments C and test whether the results survive without them (they do; Section 7.1).

Finally, two institutional signals enter as predictors. An active IMF lending arrangement is a public distress signal: countries do not enter Fund programs when financing is easy. The sovereign credit rating, compiled here as the lowest notch across S&P, Moody's and Fitch on a 1 to 21 scale, summarizes agency views of default risk. Both are ex-ante observable to the investor forming the portfolio in December.

3 Related literature

The literature on carry in currencies establishes the baseline puzzle. Lustig et al. (2011) show that currency carry returns load on common risk factors rather than diversifying away, and Menkhoff et al. (2012) link carry returns to global FX volatility. Our setting extends the carry logic from currencies to long-duration sovereign bonds, where the yield differential is larger and the default leg is explicit.

The local-currency emerging-market bond market itself is surveyed by Burger and Warnock (2006), who document the growth of these markets and the role of foreign participation. Their work treats the asset class as an investable universe; we ask whether its most basic ex-ante signal, the starting yield, predicts dollar returns.

On default, Cruces and Trebesch (2013) assemble haircut estimates across restructuring episodes and show that the NPV losses vary widely, which disciplines our choice to apply haircuts only where domestic debt was restructured and to grade those adjustments as provisional. The broader history of debt buildups and crises in Reinhart and Rogoff (2009) motivates the debt-dynamics predictor: rising debt ratios have preceded most sovereign distress episodes in the historical record.

On return predictability in bonds, Fama and Bliss (1987) show that forward rates predict excess bond returns, with the predictive power rising at longer horizons. Our finding that the starting yield's predictive power strengthens from 12 to 60 months is the frontier-sovereign analogue of that term structure.

This paper's contribution is measurement with the FX and default legs attached: a dollar-denominated return panel for frontier local-currency 10-year bonds, 18 countries, 2010 to 2024, with explicit restructuring adjustments, predictive regressions at three horizons, and the falsifiers reported rather than buried.

4 Data

4.1 Sources and definitions

The core series are the 10-year local-currency sovereign yield and the spot exchange rate (local currency per US dollar), monthly. Yields come from the IMF International Financial Statistics (FIGB_PA) for 11 countries and from the OECD harmonized long-term interest rate via FRED for 7 countries (Chile, Czechia, Hungary, Mexico, Poland, Romania, South Africa). Exchange rates come from the IMF IFS for 14 countries and from ECB euro reference rates (via cross-rate) for South Africa, Mexico, Poland and Romania. CPI series for the real-yield computation come from the same two sources. All IFS series were retrieved in October 2026 via the ElkassabgiData connector, which is a redistribution channel only; the producer citations are the authoritative ones. Evidence grades:

yields, FX and CPI are Observed, A; the ECB cross-rate FX is Reconstructed, A; returns computed from them are Modelled, B.

Annual predictors are: the December 10-year yield (*start_yield*, percent per annum); the real yield (nominal minus trailing 12-month CPI inflation); general government gross debt as a share of GDP and its 3-year change in percentage points (IMF World Economic Outlook, capped at 2024 actuals since later vintages are forecasts; Observed, A); an IMF program dummy equal to 1 if any Fund arrangement was in effect during any part of the year (hand-compiled from IMF Financial Position in the Fund pages and press releases; Reconstructed, A); total reserves in months of imports (World Bank WDI; Observed, A); the current-account balance as a share of GDP (WEO, capped 2024; Observed, A); the year-end sovereign foreign-currency rating notch, 1 to 21, lowest across S&P, Moody's and Fitch, 0 for unrated (hand-compiled from agency histories; Reconstructed, B); and trailing 12-month FX volatility (standard deviation of monthly log FX changes).

4.2 Country coverage

The universe is 18 countries: Angola, Armenia, Bangladesh, Botswana, Chile, Czechia, Fiji, Ghana, Hungary, India, Kyrgyzstan, Sri Lanka, Morocco, Moldova, Mexico, Poland, Romania and South Africa. The design target was 20 or more. It was missed for a documented, binding reason: a full probe of all 249 ISO2 codes in the IMF IFS catalogue found only 22 countries with a 10-year government bond yield series at all; after removing 8 developed markets, 1 defunct entity and 2 unusable series, 11 remained, and 7 more were added via FRED/OECD. No 10-year local-currency sovereign yield series exists in any probed source (IMF IFS, FRED, ECB, BIS, OECD catalogues) for Argentina, Turkey, Ukraine, Egypt, Nigeria, Kenya, Pakistan, Indonesia, the Philippines, Thailand, Malaysia, Colombia, Peru, Vietnam and others. The paper reports results for the covered 18 only. This is a limitation stated plainly: the frontier universe investors actually trade is larger than the universe with clean yield data.

4.3 Sample and data-quality flags

The predictability sample uses December formation dates from 2010 onward, keeps formations with starting yield at or below 35 percent, and requires the forward window to end by December 2024. That yields 220 country-year formations across the 18 countries (2010 to 2025); forward returns are available for 158 formations at 12 months, 138 at 36 months and 113 at 60 months. Summary statistics appear in Table 1.

Four data-quality flags matter for reading the results. First, Ghana's IFS yield is flat at 29.85 for 29 months (July 2022 to November 2024): post-2022 returns are carry plus FX only, with yield-change effects unobserved. Second, Sri Lanka's yield series is sparse (55 scattered months, ending April 2020) with only 24 computable monthly returns; its 2022 and 2023 default events carry flags but no return, so the DDO haircut does not enter any computed return. Third, hyperinflation-era yields (Armenia 1994 to 1998, up to 210 percent; Romania 1999 to 2001, up to 110 percent) make the modified-duration-7.5 approximation overstate price moves; the 2010 to 2024 window used for buy-and-hold figures excludes those episodes. Fourth, Mexico's FRED yield has 60 gap-months; returns are NaN across gaps by design, never interpolated. WEO debt and current-account series are capped at 2024 actuals.

Table 1. Summary statistics: annual formation sample

Variable	N	Mean	Std. dev.	Min	Max
Starting 10Y yield (% p.a.)	220	8.20	5.17	0.49	29.85
Real yield (% p.a.)	210	0.95	37.91	−255.00	245.41
Forward USD return, 12m	158	0.04	0.31	−0.79	2.76
Forward USD return, 36m	138	0.09	0.53	−0.86	4.71
Forward USD return, 60m	113	0.11	0.66	−0.81	5.72
Debt/GDP (%)	214	47.6	17.8	8.6	92.7
Debt/GDP 3Y change (pp)	214	4.2	10.0	−44.1	42.0
IMF program (share of years)	214	0.44	0.50	0	1
Reserves (months of imports)	220	5.14	1.93	1.45	12.40
Current account/GDP (%)	214	−3.5	6.0	−44.9	11.1
Rating notch (0–21)	214	10.8	4.1	0	18
FX volatility, trailing 12m	217	0.025	0.018	0.000	0.167

Notes: December formation dates, 2010 to 2025, 18 countries; starting yield capped at 35 percent. Returns are in decimals (0.04 = 4 percent). The real-yield extremes reflect hyperinflation-era CPI prints. All figures Reconstructed from the series in Appendix A.

Source: IMF IFS and WEO via ElkassabgiData; OECD via FRED; ECB; World Bank WDI; desk computations.

5 Methodology

5.1 Return computation

For each country-month with yield y (percent per annum) and FX rate F (local currency per USD):

$$R_{local,t} = \frac{y_{t-1}}{12} - 7.5\,(y_t - y_{t-1}), \tag{2}$$

with yields in decimals, and

$$R_{USD,t} = (1 + R_{local,t})\,\frac{F_{t-1}}{F_t} - 1. \tag{3}$$

The 7.5 is the assumed modified duration of a 10-year par bond. It is exact only for a par bond at moderate yields; above about 20 percent yield the true modified duration is far lower, so the approximation overstates both carry and price moves in hyperinflation and disinflation episodes. In event months where local-currency debt was restructured, the month's dollar return is reduced by the haircut: 14 percent for Ghana in February 2023, 10 percent NPV for Sri Lanka in September 2023 (the latter never enters a computed return, per Section 4.3). If yield or FX is missing in month t or $t-1$, the return is NaN; there is no interpolation across gaps. Every panel row carries data grade B: inputs are Observed, the return is Modelled.

Buy-and-hold returns compound the monthly dollar (or local) returns over each country's available window within 2010 to 2024. Because start months differ by data availability (Angola from November 2013, Fiji from March 2012, India through May 2017, Romania through June 2018), the figures are window-specific, not strictly same-period; the caption of Figure 1 states this.

5.2 Predictive regressions

For each December formation date t in country i, with forward dollar return $R^{USD}_{i,t\to t+h}$ over horizon $h \in \{12, 36, 60\}$ months, we estimate

$$R^{USD}_{i,t\to t+h} = \alpha_h + \beta_h \, start_yield_{i,t} + \gamma_h' X_{i,t} + \varepsilon_{i,t,h}, \tag{4}$$

by OLS with standard errors clustered by country. X contains the real yield, the debt/GDP level and its 3-year change, the IMF program dummy, reserves, the current-account ratio and trailing FX volatility. The coefficient of interest is β_h: the marginal association between one percentage point of starting yield and the forward dollar return. Overlapping forward windows induce serial correlation in ε; clustering by country mitigates it but does not eliminate it, so t-statistics should be read with that caveat (Section 7.2).

We also sort formations into terciles by starting yield within each formation year and report mean forward dollar returns per tercile and the high-minus-low spread, a non-parametric check that does not impose linearity.

5.3 Mechanism and falsifier specifications

The currency-wedge mechanism is tested directly: the 36-month log FX depreciation, $\log(F_{t+36}/F_t)$, regressed on the starting yield, same formation sample. Falsifiers are: (i) the 36-month slope re-estimated without the default haircuts; (ii) leave-one-out re-estimation of the 36-month slope, dropping each country in turn; (iii) the 36-month slope split by formation subperiod, 2010 to 2015 versus 2016 to 2021; and (iv) the 36-month slope re-estimated on local-currency forward returns, the hedged proxy, versus dollar returns.

This is a retrospective analysis; there was no pre-registered plan.

6 Results

6.1 Carry does not survive the currency leg

Figure 1 shows dollar buy-and-hold total returns by country, 2010 to 2024. Armenia leads at plus 441 percent (Modelled, B), followed by Morocco at plus 89 percent, Romania at plus 58 percent, Mexico at plus 55 percent and Bangladesh at plus 50 percent. At the bottom: Ghana at minus 78 percent, Angola at minus 43 percent, Kyrgyzstan at minus 13 percent, Botswana at minus 3 percent, Czechia at plus 4 percent (all Modelled, B).

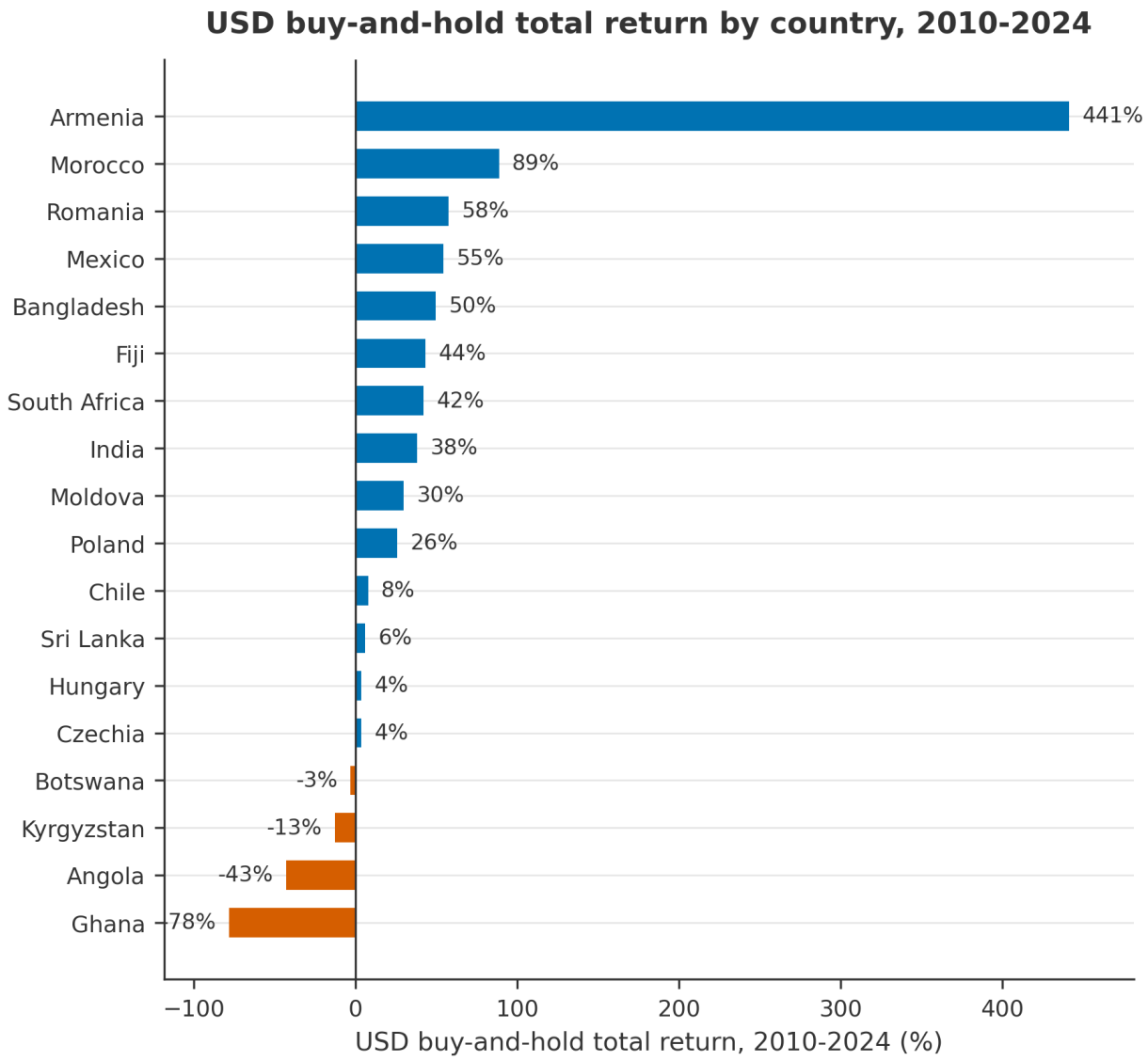


Figure 1. USD buy-and-hold total return by country. Returns compound monthly dollar total returns over each country's available window within 2010 to 2024 (start months differ by data availability; see Appendix A). Bars start at zero.

The ranking inverts the local-currency story that motivated this paper. Ghana, the plus 267 percent local-currency poster child of KR-WP-2026-03 (measured over January 2016 to September 2026), earned minus 78 percent in dollar terms over 2010 to 2024 once cedi depreciation and the February 2023 domestic debt exchange are accounted for (Modelled, B). Kyrgyzstan, the other poster child at plus 192 percent local, earned minus 13 percent in dollars (Modelled, B). The two economies that looked best in local currency sit at the bottom of the dollar ranking. Note the window difference: the plus 267 and plus 192 percent figures are KR-WP-2026-03's 2016 to 2026 modeled local-currency numbers; over this paper's 2010 to 2024 window Ghana's local-currency buy-and-hold is plus 142 percent, Kyrgyzstan's plus 91 percent. The point stands on either window: the FX leg, not the carry, decided the dollar outcome.

Armenia's plus 441 percent deserves a caution in the same sentence as the headline: it reflects

the dram's stability against the dollar combined with high local yields over the window, and the pre-2010 hyperinflation era is excluded from this figure (Section 4.3).

6.2 Starting yield predicts forward dollar returns

Table 2 reports the predictive regressions with controls. The slope on the starting yield rises with horizon: 0.024 at 12 months (t=1.33, p=0.19, R2=0.13, n=153), 0.059 at 36 months (t=1.92, p=0.055, R2=0.30, n=129), and 0.078 at 60 months (t=2.02, p=0.043, R2=0.36, n=101), with country-clustered standard errors. One percentage point of additional starting yield is associated with about 2.4 percentage points of additional 12-month dollar return and about 7.8 percentage points of additional 60-month dollar return, ceteris paribus. Figure 2 plots the three slopes with 95 percent confidence intervals: the point estimates rise monotonically while the intervals stay wide, a reminder that the premium is estimated, not measured.

Table 2. Predictive regressions: forward USD return on starting yield, with controls

	12-month	36-month	60-month
Starting yield	0.024	0.059	0.078
	(0.018)	(0.031)	(0.039)
Real yield	0.0001	−0.0003	0.0011
	(0.0003)	(0.0003)	(0.0004)
Debt/GDP	−0.001	−0.004	−0.004
	(0.001)	(0.002)	(0.003)
Debt/GDP 3Y change	−0.006	−0.015	−0.018
	(0.004)	(0.007)	(0.007)
IMF program	−0.026	−0.099	−0.30
	(0.062)	(0.156)	(0.247)
Reserves (months imports)	0.005	−0.002	−0.001
	(0.011)	(0.018)	(0.032)
Current account/GDP	0.002	0.015	0.008
	(0.004)	(0.010)	(0.020)
FX volatility, trailing 12m	1.45	14.27	17.99
	(1.30)	(4.74)	(4.54)
R2	0.13	0.30	0.36
Observations	153	129	101

Notes: OLS with standard errors clustered by country, in parentheses. Yields in percent per annum; returns in decimals. The IMF program coefficients are negative but statistically insignificant at all horizons; the audited 60-month estimates are b=-0.3012, SE=0.247, t=-1.22, p=0.223 (12m: t=-0.42; 36m: t=-0.64), per notes/results.txt. Overlapping forward windows leave residual serial correlation; see Section 7.2.

Source: Desk analysis of the panel in Appendix A.

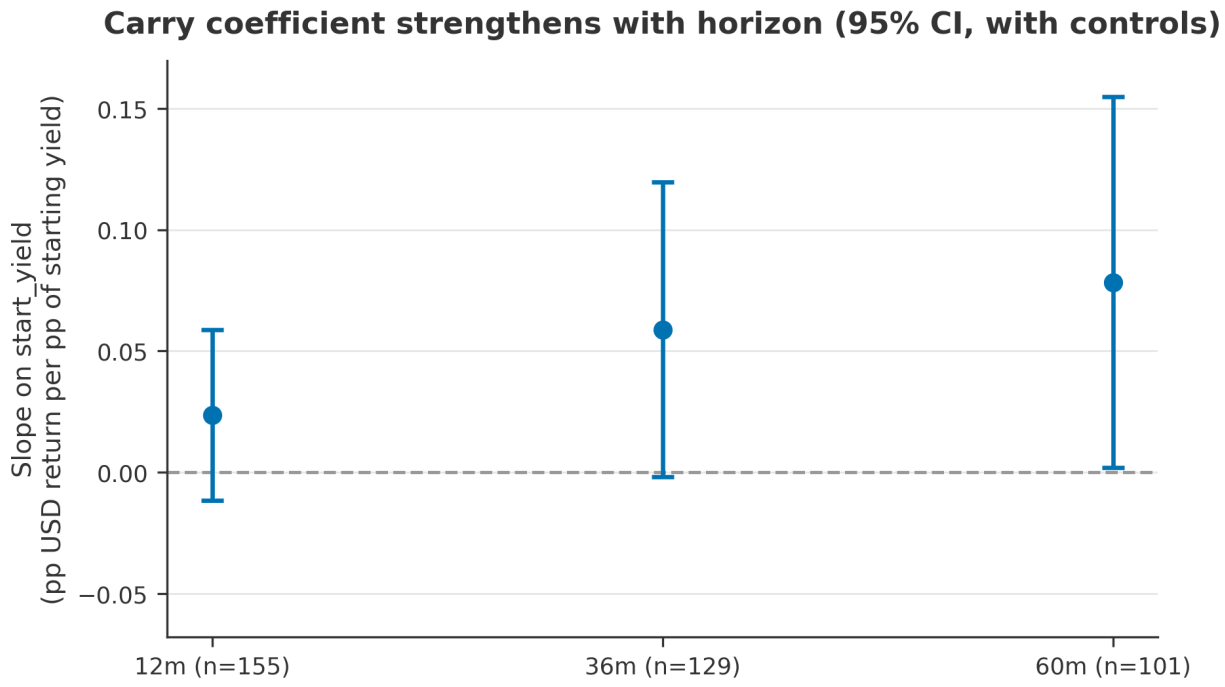


Figure 2. Carry coefficient by horizon with 95 percent confidence intervals (with-controls specifications). The point estimate rises from 0.024 at 12 months to 0.078 at 60 months; intervals are country-clustered.

The tercile sorts in Table 3 and Figure 3 confirm the pattern without imposing linearity. Sorting formations into yield terciles within each formation year, the high-minus-low forward dollar return is plus 10.9 percentage points at 12 months, plus 28.9 at 36 months and plus 35.7 at 60 months, roughly monotonic at the longer horizons.

Table 3. Tercile sorts by starting yield: mean forward USD return

Horizon	Low tercile	Mid tercile	High tercile	High − low
12 months	0.6%	−0.3%	11.5%	10.9 pp
36 months	−3.0%	−0.1%	25.9%	28.9 pp
60 months	−3.1%	7.1%	32.6%	35.7 pp

Notes: Terciles formed by starting yield within each formation year. Counts: 12m, 56/47/52; 36m, 47/41/43; 60m, 36/33/34 (low/mid/high). pp = percentage points.
Source: Desk analysis.

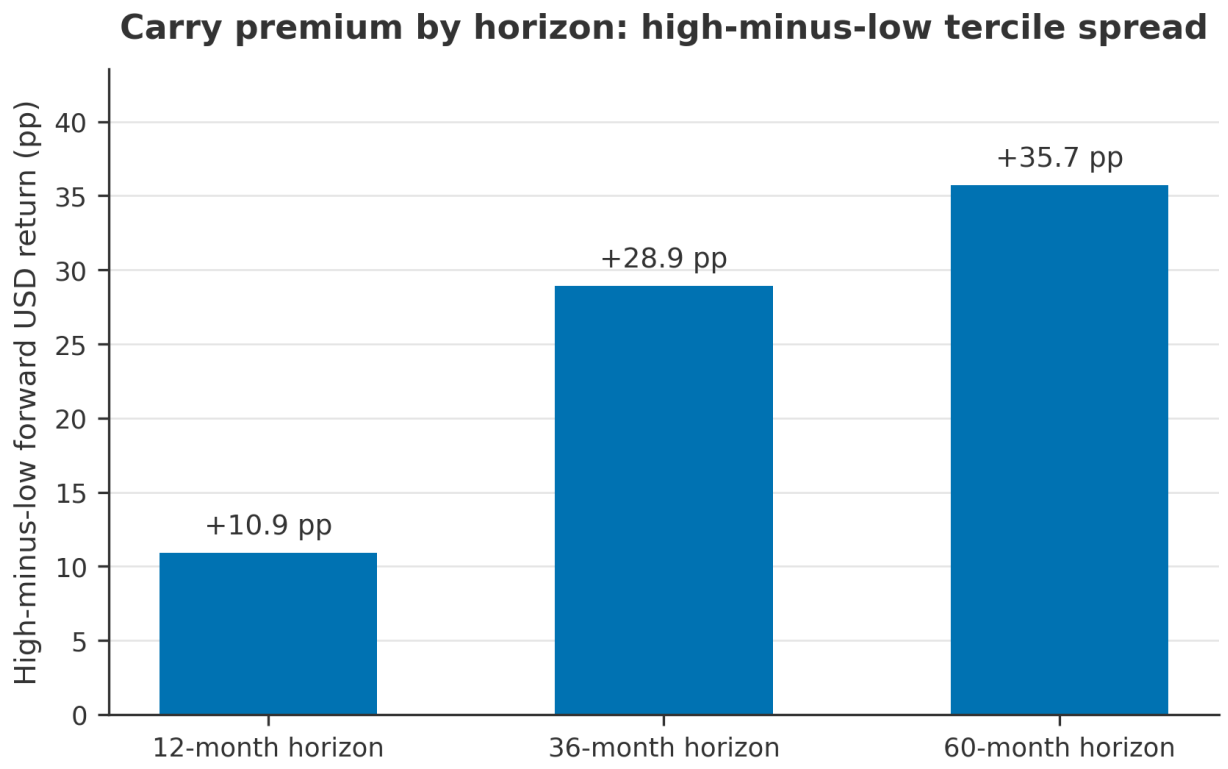


Figure 3. High-minus-low forward USD return by horizon, from within-year yield tercile sorts. The premium widens from 10.9pp at 12 months to 35.7pp at 60 months.

6.3 The mechanism is the currency wedge

Figure 4 plots each country's local-currency buy-and-hold against its dollar buy-and-hold. Points below the parity line are countries where the currency leg subtracted from local returns. The wedge is largest exactly where yields were highest: Ghana (plus 142 percent local, minus 78 percent in dollars), South Africa (plus 246 percent local, plus 42 percent in dollars), Chile (plus 111 percent local, plus 8 percent), Hungary (plus 117 percent local, plus 4 percent) and Botswana (plus 103 percent local, minus 3 percent), all Modelled, B. Armenia sits near parity (plus 439 percent local, plus 441 percent in dollars): the one high-yield market whose currency held.

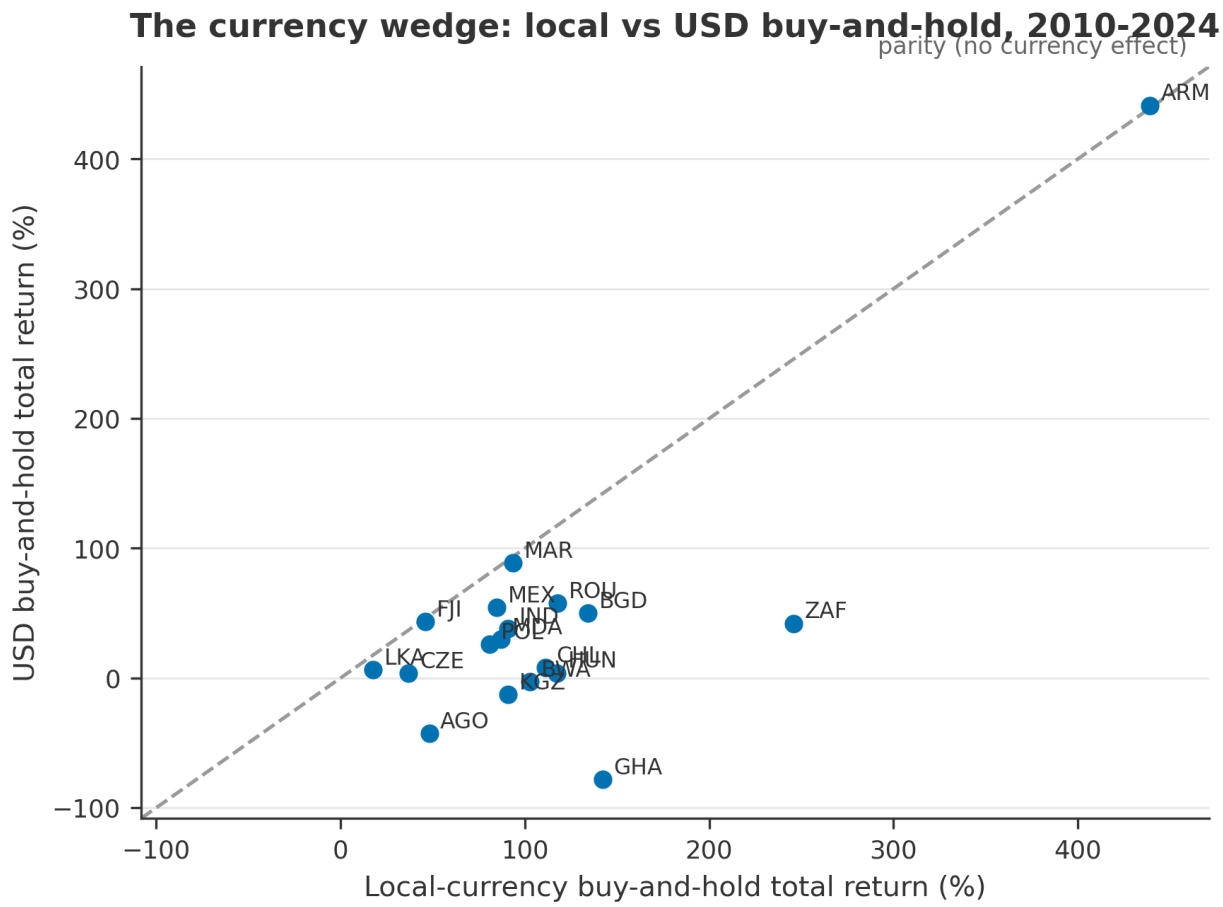


Figure 4. Local-currency vs USD buy-and-hold total return, 2010 to 2024, by country (ISO3 labels). Points below the dashed parity line lost to currency depreciation. Armenia (ARM) is the outlier near parity.

The wedge is not just a buy-and-hold curiosity; it operates formation by formation. Figure 5 scatters the starting yield against the subsequent 36-month log FX depreciation for 176 formations. The correlation is +0.245, and the regression slope is 0.011 (t=1.99, p=0.047, R2=0.06, country-clustered): each percentage point of starting yield predicts about 1.1 percent more log depreciation over the next three years. The R2 of 0.06 says depreciation is mostly noise around this relationship, which is why the carry premium survives in expectation even though the wedge eats most of it in realization. The local-versus-dollar comparison makes the same point from the other side: at 36 months the slope on the starting yield is 0.050 with t=2.89 and R2=0.32 in local currency, but 0.036 with t=1.46 and R2=0.14 in dollars. Carry is strong before FX; the currency leg takes most of it.

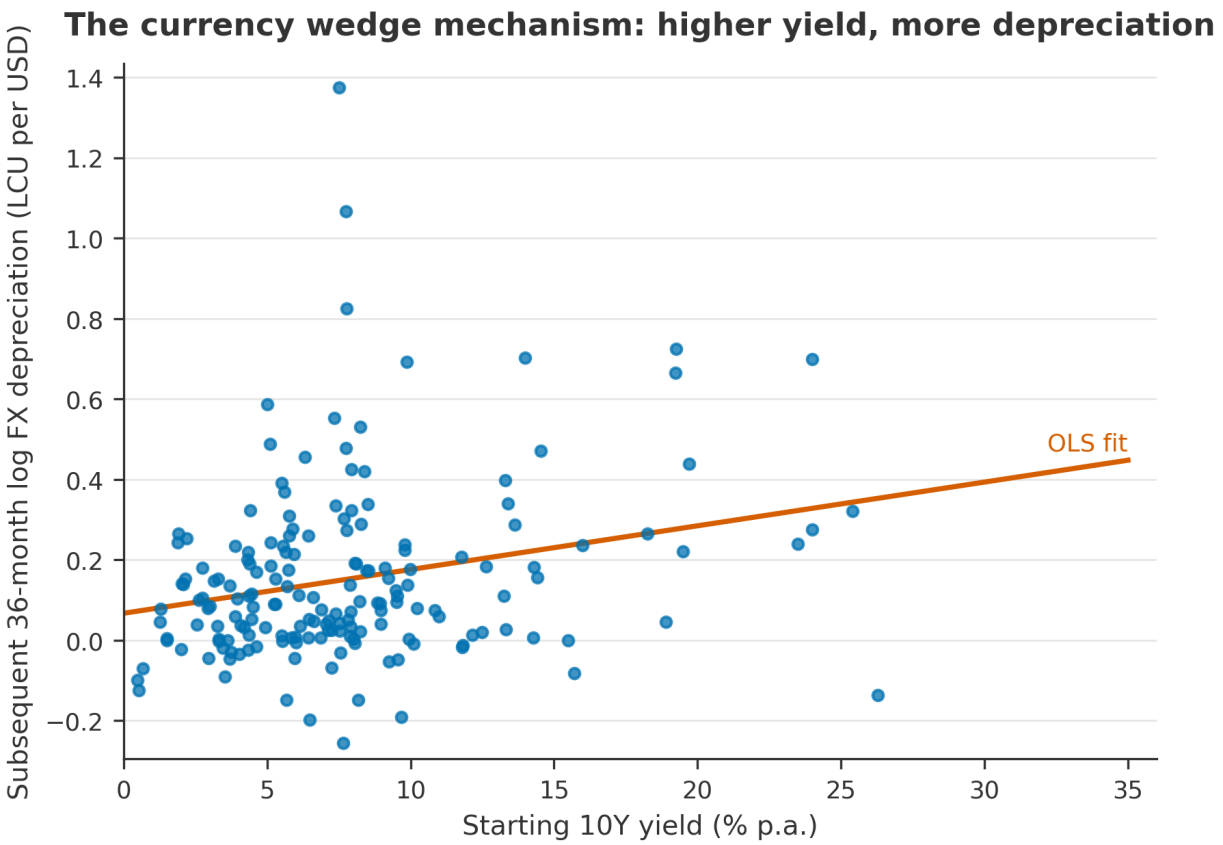


Figure 5. Starting 10Y yield vs subsequent 36-month log FX depreciation (176 December formations, 2010 onward; positive = local currency depreciation). The OLS fit has slope 0.011 per percentage point of yield (audited: t=1.99, R2=0.06, country-clustered).

6.4 The ruin leg: debt dynamics and IMF programs

In the 60-month with-controls specification, the 3-year change in the debt-to-GDP ratio predicts lower dollar returns (b=-0.018, t=-2.57, p=0.010): a 10 percentage point faster debt buildup is associated with about 18 percentage points lower 5-year dollar return. The debt effect is also significant at 36 months (t=-2.11, p=0.035), so it is the channel that carries the ruin leg. An active IMF program, the distress signal, has a negative coefficient too, but it is statistically insignificant at every horizon (12m: t=-0.42, p=0.68; 36m: t=-0.64, p=0.53; 60m: b=-0.30, t=-1.22, p=0.22). The real yield adds signal beyond the nominal yield (b=0.0011, t=2.57, p=0.010). The debt/GDP level, reserves and the current account are insignificant. Trailing FX volatility enters large and positive (t=3.96), likely a crisis-rebound effect: volatile currencies that just crashed tend to bounce. We report it as tentative, not a trading signal.

One honest qualification belongs in the same paragraph as the debt result. Figure 6 shows the raw scatter of the 3-year debt/GDP change against the 60-month forward dollar return (113 formations): the bivariate cloud tilts slightly *positive* (correlation +0.17). The negative coefficient emerges only in the multivariate specification, once the starting yield and other controls are held constant. Readers should not expect a univariate debt-buildup screen to reproduce the regression result.

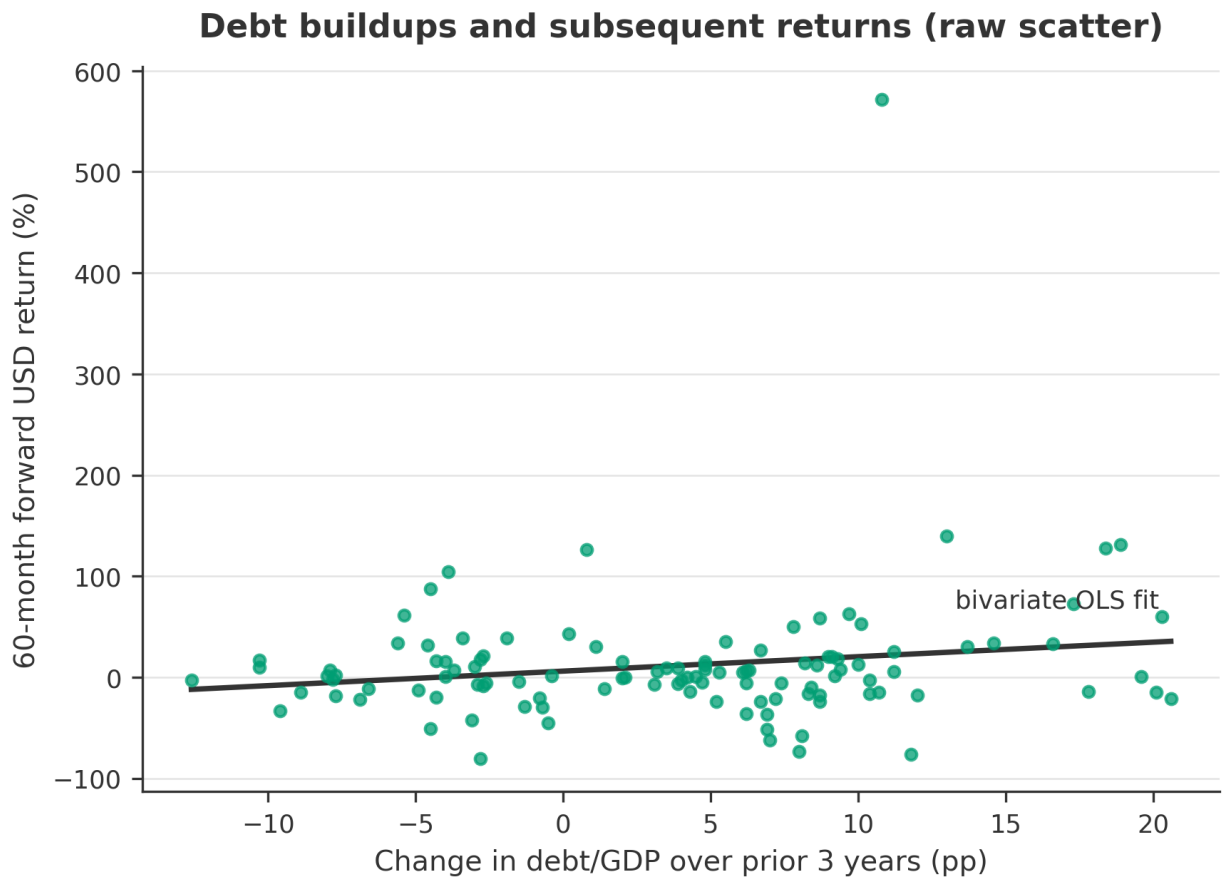


Figure 6. Change in debt/GDP over the prior 3 years vs 60-month forward USD return (113 formations; the bivariate OLS line tilts slightly positive). The negative "ruin" coefficient (b=-0.018, t=-2.57) comes from the multivariate specification in Table 2.

7 Limitations and robustness

7.1 Falsifiers

Table 4 reports the four pre-specified falsification exercises, all on the 36-month slope on the starting yield unless noted. They are the section of this paper we most want the reader to take seriously.

Panel A: the premium is not an artifact of the default adjustments. Removing the Ghana and Sri Lanka haircuts moves the 36-month slope from 0.0363 to 0.0366, a difference without economic meaning. Whatever the premium is, it is in the yields and the currencies, not in our restructuring accounting.

Panel B: two countries matter. Dropping Ghana *raises* the slope to 0.068 (t=1.81): Ghana drags the premium down, which is consistent with the buy-and-hold result but means the full-sample slope understates the premium for the rest of the universe. Dropping Moldova *halves* the slope to 0.016, though the t-statistic rises to 2.75: Moldova is influential, and the premium is weaker without it. Dropping any other country leaves the slope between 0.035 and 0.040. A result that

Table 4. Falsification exercises: 36-month slope on starting yield

Specification	Slope	t-stat	R2
A. Default adjustments			
With haircuts (baseline)	0.036	1.46	0.14
Without haircuts	0.037	1.48	0.15
B. Leave-one-out (selected)			
Full sample	0.036	1.46	n/a
Drop Ghana	0.068	1.81	n/a
Drop Moldova	0.016	2.75	n/a
Drop any other country	0.035–0.040	1.28–1.46	n/a
C. Subperiods (formation years)			
2010–2015 (n=75)	0.057	1.57	0.23
2016–2021 (n=56)	0.006	1.28	0.01
D. Currency denomination			
Local-currency forward return	0.050	2.89	0.32
USD forward return	0.036	1.46	0.14

Notes: Panels A, B and D use the no-controls 36-month specification (n=131); Panel C splits it by formation subperiod. Standard errors clustered by country.
Source: Desk analysis.

halves when one of 14 countries is removed is a result with a wide confidence interval around its economic interpretation, whatever the t-statistic says.

Panel C is the sharpest falsifier. The premium is a pre-2016 phenomenon: 2010 to 2015 formations show a slope of 0.057 with R2=0.23, while 2016 to 2021 formations show a slope of 0.006 with R2=0.01. There is no out-of-sample stability. Any investor-facing reading of this paper must carry that sentence.

Panel D restates the mechanism as a falsifier of the naive carry story: the slope is 0.050 with t=2.89 in local currency and 0.036 with t=1.46 in dollars. An investor who could hedge cheaply would face a different opportunity set; Section 2 explains why cheap hedging is not available.

7.2 Limitations

Overlapping forward windows are the main statistical limitation. A 60-month forward return formed in December 2010 overlaps 59 of the 60 months of the December 2011 formation's window. Country clustering mitigates the resulting serial correlation but does not eliminate it; the reported t-statistics are best read as upper bounds on precision.

The duration approximation (modified duration 7.5 for all bonds in all months) overstates price moves at high yields and misstates them where Ghana's yield went stale. The return panel is Modelled, B throughout, and the 2010 to 2024 buy-and-hold window was chosen precisely to exclude the hyperinflation-era episodes where the approximation breaks down.

Coverage is 18 countries, not the 20-plus the design targeted, and the missing economies (Argentina, Turkey, Ukraine, Egypt, Nigeria, Kenya, Pakistan and others) are missing because no clean 10-year

local-currency yield series exists for them, not because they were screened out. Sri Lanka's 24 return observations cannot support inference about its restructuring. The IMF program dummy mixes precautionary arrangements (Mexico's undrawn Flexible Credit Line, Chile's) with active distress programs, which attenuates its interpretation as a pure distress signal; in any case its coefficient is statistically insignificant, so the data do not support a distress-signal claim for the dummy. The FX-volatility finding is reported as tentative: a large positive coefficient on trailing volatility smells of crisis rebound, not a risk premium an investor can harvest ex ante.

Falsifiability. Three observations would weaken or overturn the main conclusions. First, if starting yields from 2022 to 2025 formations fail to predict 2026 to 2030 dollar returns, the pre-2016-only finding extends to the full sample and the premium is historical, not structural. Second, if high-yield frontier currencies systematically appreciate rather than depreciate over 36-month windows, the currency-wedge mechanism fails and the local-versus-dollar gap closes from the other side. Third, if debt buildups stop predicting lower returns in the multivariate specification as WEO vintages revise, the ruin leg was a sample artifact. This is a retrospective analysis; there was no pre-registered plan.

8 Conclusion

Rebuilt in dollar terms with explicit default adjustments, the frontier sovereign carry story splits in two. As a buy-and-hold proposition, carry did not survive the currency leg: Ghana's plus 267 percent local-currency figure became minus 78 percent in dollars, and the two poster children of the local-currency ranking sit at the bottom of the dollar ranking. As a cross-sectional predictor, the starting yield did survive: it predicts forward dollar returns at 12, 36 and 60 months, more strongly at longer horizons, and the high-yield tercile beat the low-yield tercile by 35.7 percentage points over five years. The two findings reconcile through the currency wedge: higher yields predict more depreciation, so carry pays in expectation while the FX leg eats most of it in realization. The ruin leg is the qualifier: rising debt ratios predict lower dollar returns, the IMF dummy is negative but insignificant, and the premium itself is pre-2016, Moldova-sensitive, and Ghana-dragged. An investor who wants the carry must underwrite the currency; the yield is not a free lunch, it is the market's estimate of the lunch's price.

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A Data appendix

Table 5 inventories every input series with its producer citation, retrieval date and data vintage. The build script `data/build_panel.py` reconstructs `panel_monthly.csv` (11,348 rows) and `predictors_annual.csv` (407 rows) from the raw files in `data/raw_*/` plus the reused IFS yields and the two hand-compiled working files (`ratings_working.csv`, `imf_program_working.csv`). Full per-country coverage windows and the exclusion list are documented in `data/coverage_report.md`.

Table 5. Series inventory

Series	Producer (cite as)	Retrieved	Data through
10Y yield, 11 countries	IMF IFS (FIGB_PA) via ElkassabgiData	2026-10-08	2025-04/06
10Y yield, 7 countries	OECD via FRED (IRLTLT01)	2026-10-08	2026-07/08
Spot FX, 14 countries	IMF IFS (ENDE_XDC_USD_RATE)	2026-10-08	2025-06/07
Spot FX, 4 countries	ECB euro reference rates (cross-rate)	2026-10-08	2026-09
CPI, 14 countries	IMF IFS (PCPI_IX)	2026-10-08	2025-06/07
CPI, 4 countries	OECD via FRED (CPALTT01)	2026-10-08	2024-07 to 2025-03
Govt. debt, % GDP	IMF WEO (GGXWDG_NGDP)	2026-10-08	2024 (actuals)
Current account, % GDP	IMF WEO (BCA_NGDPD)	2026-10-08	2024 (actuals)
Reserves, months imports	World Bank WDI	2026-10-08	2024
Sovereign rating notch	S&P, Moody's, Fitch via countryeconomy.com	2026-10-08	2010–2024
IMF program dummy	IMF Financial Position in the Fund	2026-10-08	2010–2024

Notes: ElkassabgiData is a redistribution channel only; producer citations are authoritative. WEO series are capped at 2024 actuals; later vintages are forecasts. Grades: yields/FX/CPI/debt/reserves Observed, A; ECB cross-rate FX Reconstructed, A; ratings Reconstructed, B; IMF programs Reconstructed, A; computed returns Modelled, B; haircuts Modelled, C.
Source: Desk data appendix (data/data_appendix.md).